

Natural Language Processing (NLP)

MPhil / MS Data Science

ClearClause

AI-Powered Contract Risk Analyzer

for Freelancers and Professionals

Project Deliverable 4

Business (Y Combinator) and Technical Perspective

Student ID	Name	Role
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Project Snapshot

One-line YC pitch: ClearClause helps freelancers, employees, and small businesses understand risky contracts in under 60 seconds by extracting legal clauses, scoring risk, rewriting legalese into plain English, and answering document-specific questions through grounded NLP.

Project Name	ClearClause
Category	AI LegalTech / B2C-to-B2B SaaS
Primary Users	Freelancers, fresh graduates, employees, small businesses, renters, independent professionals
Core NLP Tasks	Clause extraction, risk classification, simplification, summarization, RAG-based Q&A
Prototype Stack	Python, Streamlit, pdfplumber, python-docx, Groq LLaMA3, LangChain, FAISS, sentence-transformers, Plotly

1 Project Description

1.1 Problem

Freelancers, fresh graduates, small businesses, and independent professionals sign contracts every day without fully understanding the obligations hidden inside them. Employment agreements, NDAs, freelance service contracts, vendor agreements, leases, and platform terms often contain dense legal wording around intellectual property, termination, liability, payment, arbitration, non-compete restrictions, and automatic renewal.

The problem is not only legal complexity; it is information asymmetry. A company or client can draft a contract with legal support, while the signer often has no affordable way to identify what is risky before accepting it. Hiring a lawyer for every small freelance or employment contract is unrealistic, especially in emerging markets and early-career professional communities.

1.2 Solution

ClearClause is an NLP-powered contract review system that turns a legal document into a practical risk report. A user uploads a PDF, DOCX, or TXT contract, and the system:

1. Extracts contract text using document parsers.
2. Detects important legal clause types.
3. Classifies each clause as high, medium, or low risk.
4. Simplifies complex legal language into plain English.
5. Produces a structured document summary.
6. Builds a RAG index so the user can ask questions grounded in the uploaded contract.
7. Generates a 0–100 risk score and an exportable analysis report.

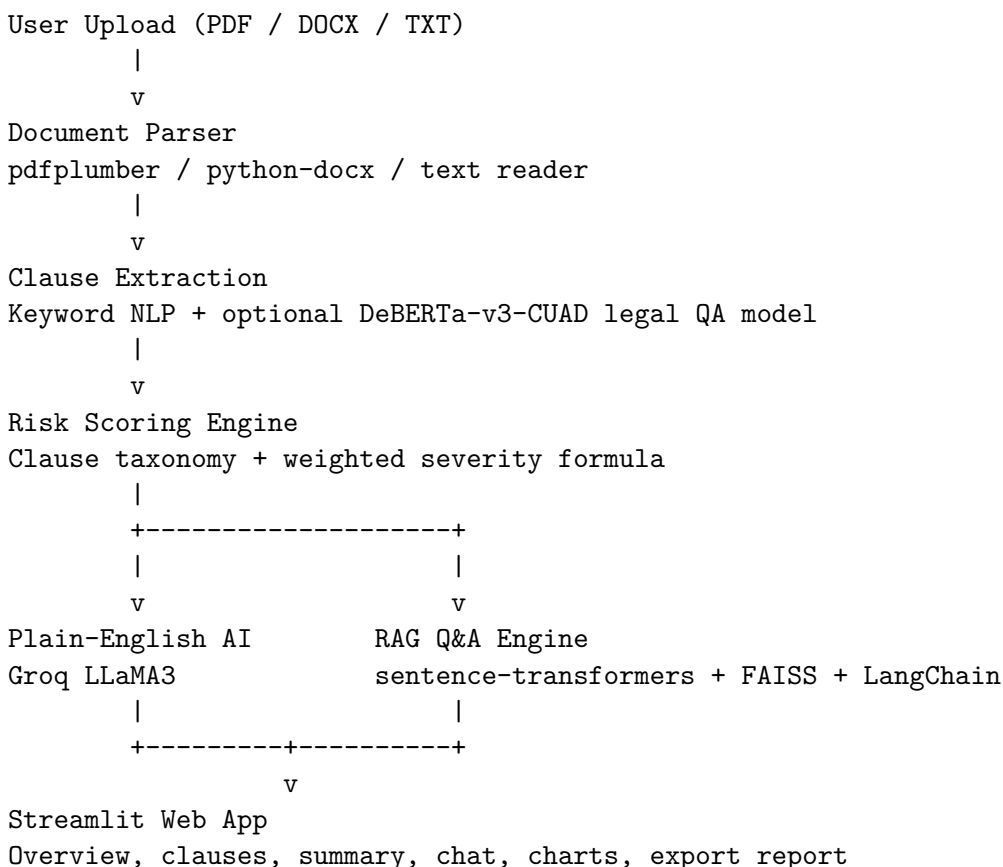
The output is not positioned as a replacement for a lawyer. It is an early-warning and comprehension layer that helps users know what to inspect, what to negotiate, and when to seek professional legal advice.

1.3 Target Users

User Segment	Pain Point	ClearClause Benefit
Freelancers and gig workers	Client contracts contain IP transfer, payment, and liability risks	Fast plain-English risk scan before signing
Fresh graduates and employees	Employment terms are difficult to interpret	Highlights non-compete, probation, termination, and IP assignment clauses
Small businesses and startups	Vendor/SaaS agreements are reviewed without legal staff	Affordable first-pass contract intelligence
Renters and independent professionals	Lease and service agreements contain penalties and renewal traps	Converts hidden obligations into readable summaries

2 Technical Perspective

2.1 Current System Architecture



2.2 Project Tech Stack

Component	Technology	Reason for Selection
Document Parsing	pdfplumber, python-docx, TXT parser	Supports common contract formats while keeping ingestion local and lightweight.

Component	Technology	Reason for Selection
Clause Extraction	Keyword NLP with optional tomasonjo/deberta-v3-base-cuad	Keyword extraction keeps the prototype fast; the CUAD-trained DeBERTa model provides a path toward legal-domain extractive QA.
Risk Classification	Rule-based legal clause taxonomy	Transparent for academic evaluation and explainable for users.
LLM Layer	Groq API with LLaMA3-8B	Fast and affordable inference for simplification, summarization, and grounded answers at demo scale.
Embeddings	sentence-transformers/all-MiniLM-L6-v2	Efficient semantic embeddings for document chunks without heavy infrastructure.
Vector Store	FAISS	In-memory vector retrieval keeps private document context temporary and avoids database setup for the prototype.
RAG Framework	LangChain	Standard retrieval orchestration for contract-specific Q&A.
Frontend	Streamlit	Rapid Python-native UI for upload, charts, clause review, summaries, and chat.
Visualization	Plotly	Interactive risk gauge and chart breakdowns.
Configuration	.env + python-dotenv	Keeps API keys out of source code and supports simple local setup.
Future SaaS API	FastAPI, Docker, Azure	Clear path from academic prototype to scalable web/API product.

2.3 Core NLP Modules

2.3.1 Module 1 - Document Ingestion

- Input: PDF, DOCX, or TXT contract.
- Process: extract raw text, normalize whitespace, split into paragraphs/chunks.
- Output: clean text corpus for extraction, summarization, and retrieval.

2.3.2 Module 2 - Clause Extraction and Risk Detection

- Current implementation: keyword-based detection across 14 legal clause categories.
- Upgrade path: extractive QA using a CUAD-trained DeBERTa-v3 model.
- Example high-risk clauses: unlimited liability, automatic renewal, unilateral termination, non-compete, indemnification, IP ownership/work-for-hire.
- Output: structured clause records with category, risk level, evidence text, and explanation.

2.3.3 Module 3 - Plain-English Simplification

Instruction prompting through Groq LLaMA3 rewrites legal clauses into one or two non-lawyer-friendly sentences without changing meaning. This is a product-critical layer because users need practical comprehension, not only legal labels.

2.3.4 Module 4 - RAG-Based Contract Q&A

The uploaded contract is chunked, embedded with all-MiniLM-L6-v2, stored in an in-memory FAISS index, and queried through LangChain. The intended guardrail is that answers should be based only on retrieved contract chunks.

2.3.5 Module 5 - Summarization

The system produces a structured report covering obligations, rights, risks, money/payment terms, and important dates. This gives users a quick overview before they inspect individual clauses.

2.3.6 Module 6 - Risk Scoring

Risk Score = (High clauses x 30 + Medium clauses x 15 + Low clauses x 5)
normalized and capped at 100

Score Range	Label	User Meaning
0–33	Low Risk	Mostly standard; still review before signing.
34–66	Review Carefully	Several clauses deserve attention or negotiation.
67–100	High Risk	Strong warning; consider legal advice before signing.

2.4 Evaluation Plan

Module	Metric	Evaluation Method
Clause Extraction	Precision, Recall, F1	Compare detections against manually labeled sample contracts or CUAD-style annotations.
Risk Classification	Accuracy, Macro-F1	Evaluate severity labels against a human-created risk taxonomy.
Simplification	Readability improvement, human rating	Compare original clauses with simplified outputs.
Summarization	ROUGE-1/2/L and usefulness rating	Compare generated summaries with reference summaries.
RAG Q&A	Faithfulness, source relevance, BERTScore	Check whether answers are grounded in retrieved chunks.
Product Performance	Latency, upload success rate	Measure time from upload to full report generation.

3 Business Perspective - YC Application Fields

3.1 What Are You Making?

ClearClause is a legal-document comprehension product for people who cannot afford repeated legal review. The first version focuses on freelance, employment, NDA, vendor, and service contracts. The product converts a contract into a risk dashboard, plain-English explanations, and a contract-specific chatbot.

3.2 Founder Roles

- **Shehroz Ali - Founder:** Owns NLP pipeline design, clause extraction, risk scoring, model integration, and evaluation.

- **Arslan Ahmed - Co-founder:** Owns system design, Streamlit demo experience, report interface, business model, and go-to-market planning.

3.3 Category

AI LegalTech / B2C-to-B2B SaaS. ClearClause begins as a self-serve tool for freelancers and professionals, then expands into B2B APIs for freelance platforms, HR software, universities, startup incubators, and small-business service providers.

3.4 Domain Expertise

The team combines NLP implementation experience, RAG pipeline design, legal-domain transformer research, and a working prototype. The project is grounded in real contract-review workflows: clause extraction, risk explanation, summarization, and user Q&A.

3.5 Why Now?

1. LLM APIs make legal-language simplification fast enough for consumer workflows.
2. Open legal NLP datasets such as CUAD reduce the need for expensive from-scratch labeling.
3. RAG makes document-grounded answers more reliable than generic chatbot responses.
4. Freelance and remote work continue to normalize cross-border contracts.
5. Enterprise legal AI tools are focused on law firms and large companies, leaving an affordability gap.

3.6 Market Opportunity

External market research estimates the global legal AI market at approximately USD 1.45B in 2024, projected to approach USD 3.90B by 2030 with a 17.3 percent CAGR. Broader legal technology reports estimate a much larger legal tech market, with one report projecting USD 55B by 2029.

ClearClause does not need to win the enterprise legal market first. Its beachhead is the underserved contract-comprehension market for freelancers, students, early-career professionals, and small businesses. This segment is large, price-sensitive, and poorly served by enterprise-first legal AI platforms.

3.7 Competitors

Competitor	Type	Limitation	ClearClause Edge
Harvey AI	Enterprise legal AI	Built for law firms and legal departments, not freelancers	Affordable, self-serve, contract-first workflow
LawGeex / Kira	Enterprise review	Procurement-heavy and designed for corporate legal teams	Lightweight upload-to-report experience
ChatGPT / generic LLMs	General AI assistant	Hallucination risk and no fixed clause taxonomy	RAG-grounded answers plus explicit clause categories
Manual lawyer review	Professional service	High cost for small contracts	First-pass triage before paid legal help
Manual reading	No tool	Slow and inconsistent for non-lawyers	60-second risk report with plain-English explanations

3.8 Revenue Process and Offering

Freemium lead magnet: Free users can upload a limited number of documents per month and see the overall risk score, summary, and number of flagged clauses. Paid users receive full clause-level explanations, chat, and export reports.

Tier	Price	Offering
Free	USD 0	Limited documents, basic summary, basic risk score
Pro	USD 9.99/month	Unlimited documents, full AI explanations, Q&A, export reports
Business	USD 49/month	Team workspace, bulk review, branded reports, priority usage
Enterprise/API	Custom	API access, HR/freelance platform integrations, admin controls

Additional revenue streams include legal referral marketplace fees for high-risk contracts, B2B API integrations, and university/incubator licensing.

3.9 Go-To-Market Plan

1. Launch with Pakistan-based freelancer and student communities.
2. Publish contract teardown content showing real examples of risky clauses.
3. Partner with university entrepreneurship centers and freelancing bootcamps.
4. Offer free public demos for Fiverr/Upwork-style agreements and NDAs.
5. Convert frequent users to Pro and small agencies to Business.
6. Use anonymized clause-frequency analytics to improve product positioning.

3.10 Moat

- Domain-specific clause taxonomy for freelancer and employment contracts.
- Human feedback loop from user corrections and high-risk reports.
- Growing dataset of anonymized clause patterns by contract type.
- RAG-first architecture that keeps answers grounded in uploaded text.
- Regional positioning for South Asian freelancers before expanding globally.

3.11 Investment and Fundraising

The project is currently bootstrapped as an academic prototype. A YC-style pre-seed round would be used for legal expert review, higher-quality annotated evaluation data, secure document infrastructure, API and inference costs, onboarding, user acquisition experiments, and B2B integrations.

3.12 Risks and Mitigation

Risk	Mitigation
Users mistake the product for legal advice	Clear disclaimers, not-legal-advice positioning, referral flow for high-risk cases
LLM hallucination	RAG grounding, retrieved source snippets, conservative prompts, no answer when context is insufficient
Incorrect risk classification	Transparent rule taxonomy, human evaluation, legal expert review
Sensitive document privacy	Local parsing, temporary in-memory FAISS index, secure SaaS roadmap
Low willingness to pay	Freemium funnel, affordable pricing, B2B channels, legal referral revenue

4 Project Deliverables and Launch Roadmap

4.1 Current Prototype Deliverables

- Streamlit app for upload and interactive analysis.
- PDF/DOCX/TXT document parser.
- Clause extraction module across 14 clause categories.
- Risk scoring and Plotly visualization.
- Groq-powered simplification and summarization.
- LangChain + FAISS RAG chatbot.
- Sample contracts and README setup instructions.
- Exportable analysis report.

4.2 Eight-Week Launch Roadmap

Timeline	Milestone	Output
Week 2	Evaluation Dataset MVP	20–30 manually labeled contracts across freelance, NDA, employment, and service categories
Week 4	Clause Intelligence Upgrade	CUAD-based DeBERTa QA option integrated and benchmarked against keyword baseline
Week 6	Product Hardening	Better prompts, source snippets, privacy disclaimer, improved UI, exportable PDF report
Week 8	Pilot Launch	Pilot with 25–50 freelancers/students, collect feedback, measure conversion and accuracy

4.3 Academic Submission Artifacts

- Source code repository.
- Technical and YC-style business proposal.
- Working demo application.

- Evaluation plan and sample evaluation results.
- Final presentation/demo video.

5 Success Metrics

5.1 Technical Metrics

Metric	Target
Clause extraction F1	0.80+ on project evaluation set
Risk label accuracy	0.85+ on manually labeled clauses
Average report generation time	Under 60 seconds for standard contracts
RAG answer faithfulness	0.85+ human-rated grounded answers
Simplification readability	At least 30 percent reduction in reading complexity

5.2 Business Metrics

Metric	Target
Activation	60 percent of visitors upload or load a sample contract
Free-to-paid conversion	5–8 percent in early freelancer segment
Retention	30 percent monthly active return rate among Pro users
Referral trigger	10 percent of high-risk reports create legal referral interest
B2B pilots	3 pilot partners in universities, agencies, or freelancer communities

6 Conclusion

ClearClause turns contract review from a confusing, expensive, and intimidating process into a fast NLP workflow that ordinary professionals can use before signing. Technically, the project demonstrates document parsing, legal clause extraction, risk classification, LLM simplification, summarization, RAG-based Q&A, and interactive visualization in a single application.

From a YC perspective, the product attacks a clear affordability gap: enterprise legal AI is advancing quickly, but freelancers and small teams still lack a practical tool for everyday contracts. By starting with a narrow, painful workflow and expanding toward platform integrations, ClearClause has a credible path from academic prototype to useful LegalTech SaaS.

Sources Consulted

- GuardPoint AI - YC Strategic Proposal PDF supplied as the structural reference.
- Grand View Research, Legal AI Market Size and Outlook: <https://www.grandviewresearch.com/horizon/outlook/legal-ai-market-size/global>
- Grand View Research, Legal AI Market Report: <https://www.grandviewresearch.com/industry-analysis/legal-ai-market-report>
- Arizton, Legal Tech Market Report: <https://www.arizton.com/market-reports/legal-tech-market>
- Harvey AI official website: <https://www.harvey.ai/>